

AraScholar: Calibrating Academic Paraphrasing in Modern Standard Arabic

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Abstract

Academic paraphrasing in Modern Standard Arabic must change expression without distorting scientific content. We study this balance through AraScholar, a corpus of 20,002 manually written source–paraphrase pairs with 522 raw domain labels and item-level provenance. On 2,000 held-out sentences, fine-tuned AraT5v2 favors lexical reuse under beam search, while frontier models produce more extensive rewrites. With corrected Arabic punctuation normalization, diverse decoding and preservation-constrained reranking attain 74.25% coverage of a fixed novelty–preservation band, compared with 52.60% for zero-shot GPT-4o. Gold coverage is only 46.65%, showing that this band measures constraint compliance rather than human quality. Post-hoc target matching substantially improves a calibrated GPT-4o control and narrows the gap, although candidate budgets differ. Crucially, an independent 200-instance audit favors frontier models on meaning and paraphrase quality. We document the corpus, decoding controls, negative distillation result, and metric sensitivity, including test-derived calibration and checkpoint-selection limitations. The findings establish operating-point control as a useful capability while demonstrating why calibration must be evaluated alongside independent assessments of writing quality.

1 Introduction

Academic rewriting must change expression while preserving claims, qualifications, and technical terms. When the source is already scholarly prose, correction or connective substitution alone is insufficient. In Modern Standard Arabic (MSA), morphological and syntactic variation further complicates assessment with a single automatic score.

Two contrasting failures motivate *operating-point calibration*: source repetition with high semantic similarity, and fluent reformulation that

changes the claim. We seek joint control of novelty and preservation under an annotation standard. Calibration here concerns rewriting behavior, not model confidence probabilities.

AraScholar provides 20,002 manually constructed academic source–paraphrase pairs. Four native Arabic experts performed collection, rewriting, linguistic review, and quality control. Each pair retains attribution to the quoted work and the document containing the quotation. The corpus has 522 raw domain labels, of which 513 occur in training. Its guidelines favor meaning-preserving, concise reformulation, but the observed distribution contains substantial variation. Accordingly, its mean behavior describes one annotation standard rather than a universal optimum for Arabic writing.

We compare decoding controls and reward-ranked fine-tuning of AraT5v2 (~300M parameters). Sampling, reranking, and a soft copy penalty counter beam-search reuse; the tested distillation does not. Frontier baselines instead tend toward more lexical change and longer outputs.

The evaluation yields two qualifications that are central to our conclusions. First, reranking optimizes measurements also used by the automatic evaluation, so higher coverage cannot establish independent quality superiority. Second, after correcting Arabic punctuation normalization, only 46.65% of gold rewrites satisfy the original fixed band. A selector can concentrate its outputs inside that region more often than humans while missing important properties of human writing.

We therefore examine gold-derived bands, held-out preservation encoders, distributional distance, calibrated frontier controls, and human ratings. A symmetric novelty-targeting rule improves both generators in a post-hoc control, narrowing the apparent small-model advantage. The independent audit reveals a clearer quality difference: frontier outputs receive higher ratings for meaning and

paraphrase quality. An automatic fluency constraint reduces one failure proxy, but its human benefit has not been evaluated.

Our contributions are:

- An expert-created MSA academic paraphrasing corpus with documented construction, item-level provenance, corrected domain accounting, and a controlled-access source-text policy.
- A joint novelty–preservation framework with corrected lexical measurement and explicit separation of fixed-band compliance, distributional resemblance, and human quality.
- An empirical analysis of decoding controls, calibrated frontier-model comparisons, and negative distillation results, including the limits introduced by test-derived targets and shared evaluation metrics.

The resulting claim concerns controllable rewriting. It does not equate a compact model’s calibration advantage with superior academic prose.

2 Related Work

Style transfer and multilingual editing.

Instruction-based rewriting allows models to express stylistic changes without task-specific adaptation. Reif et al. (2022) demonstrate augmented zero-shot style transfer, while mEdIT extends instruction tuning to multilingual grammatical error correction, simplification, and paraphrasing (Raheja et al., 2024). These tasks overlap with academic rewriting but do not specify how much change a scientific sentence should undergo. We discuss mEdIT as related work without claiming results for it on AraScholar.

XFORMAL evaluates formality transfer in Brazilian Portuguese, French, and Italian (Briakou et al., 2021). AraScholar starts from academic quotations rather than informal text. Its task is therefore scholarly reformulation; register elevation is not required in every example, and formality alone does not certify preservation of a scientific claim.

Novelty control and candidate selection.

Chowdhury et al. (2022) introduce novelty-conditioned retrieval-augmented prompt tuning. We build on the broader motivation for controlling lexical change, examining its interaction with preservation and human quality in Arabic. Our

distinction between novelty maximization and symmetric target matching matters when generators have different failure modes: increasing novelty can help a copier but harm an already expansive generator.

AraT5 establishes Arabic text-to-text modeling (Nagoudi et al., 2022); AraT5v2 is described within Octopus (Elmadany et al., 2023). We fine-tune an existing AraT5v2 checkpoint rather than pretrain a new generator. Nucleus sampling explores alternatives to a high-probability beam output (Holtzman et al., 2020), while reward-ranked fine-tuning converts selected outputs into supervision (Dong et al., 2023). Our negative result concerns one such configuration, not the general viability of distillation.

Evaluation of rewriting. BLEU, chrF++, and ROUGE measure overlap with a reference (Papineni et al., 2002; Popović, 2017; Lin, 2004). Sentence encoders such as LaBSE and GATE provide complementary semantic comparisons (Feng et al., 2022; Nacar et al., 2025). Neither perspective alone establishes acceptable paraphrasing: a single reference cannot cover every legitimate expression, and high sentence similarity can miss a changed qualifier or disfluency. We retain these measurements but interpret them alongside lexical change, explicit controls, and an independent human audit.

3 The AraScholar Corpus

3.1 Source selection and provenance

AraScholar contains 20,002 pairs (S, T) , where S is a quotation extracted from academic writing and T is a manual MSA paraphrase. Sources include printed academic books, refereed journal articles, and master’s and doctoral theses. The collection excludes the Qur’an, Hadith, and poetry. It targets academic quotations, not a random sample of all Arabic sentences or disciplines.

Collectors located quotations identified by quotation marks, author–year citations, or numbered footnotes. They extracted one complete, self-contained sentence and recorded its identifier and bibliographic context. Two provenance fields are nonempty in all 20,002 records: the citation of the *quoted work* and the *containing document* where the quotation was encountered. They contain 18,752 and 2,383 distinct strings, respectively. Distinct citation strings are not necessarily distinct

works: different pages or citation formats can refer to the same publication.

Quoted-work citations record author, title, publisher or journal, edition, year, and page. Sources average 29.3 words (median 26; P95 60; maximum 159). Attribution supports traceability but does not establish redistribution rights.

3.2 Manual rewriting and quality control

The team comprised four native Arabic speakers with academic-writing expertise, one responsible for each of collection, rewriting, linguistic review, and quality control. No LLM, machine translation system, or paraphrasing tool assisted the construction. The written methodology specifies five criteria: retain the quotation’s context and meaning, change its wording, maintain cohesion and appropriate style, ensure grammatical and orthographic correctness, and keep the rewrite no longer than the quotation.

Linguistic review checks unique numbering, duplication, correctness of the extracted source, soundness of the rewrite, lexical difference with semantic agreement, and correctness of the rewritten text. Errors in printed source passages could be corrected during construction. Thus, the stored source is an editorially reviewed sentence, not necessarily a byte-for-byte transcription. A separate quality stage checks domain coverage, attribution, sentence length, and lexical overlap.

The length ceiling is a guideline rather than an empirically universal constraint: 61.8% of targets are no longer than their sources, while 38.2% exceed them. The corpus-wide target/source ratio has mean 0.990 and median 0.960. These statistics characterize a broadly concise corpus but do not justify declaring every expanded system output invalid. Mean sentence-level type–token ratio increases from 0.931 to 0.954, indicating greater lexical diversity without independently proving improved register.

The four-role review process is not a multiple-annotation agreement experiment. Individual degrees, recruitment and compensation arrangements, pilot procedures, rejection/adjudication rates, and double-annotation statistics are not reported. The output-quality evaluator described in Section 7.3 was independent of both corpus construction and model development.

Partition	Pairs	Raw labels
Training	16,002	513
Development	2,000	316
Test	2,000	317
Full corpus	20,002	522

Table 1: Fixed partitions and domain counts. The corpus has 495 labels after mechanical definite-article stripping; this is not a curated disciplinary taxonomy.

3.3 Splits and domain accounting

The fixed split is shown in Table 1. Exact source-string checks found no development or test source in training and no identity pair $S = T$. This verifies exact source disjointness, not separation by document, author, or semantic content. Related quotations or publications can still span partitions.

The earlier count of 513 applies only to training. Across the corpus there are 522 raw labels and 495 after stripping the definite article ال. This mechanical operation merges spelling variants such as *اللسانيات* and *لسانيات*; it does not consolidate related categories such as *علم اجتماع* and *اجتماع*. Corpus-wide median frequency is 9.5 pairs per raw label, with 51 singletons; training has median 8.0 and 62 singletons. Appendix A reports leading normalized labels. We make no delivered-data claim about the collection plan’s separate broad-field taxonomy, for which no item-level mapping exists.

The long tail limits domain-level inference: we analyze subgroups only when they contain at least ten test examples, without claiming balanced coverage.

4 Task and Evaluation

A system produces P from source S without seeing the paired target T . References characterize the annotation distribution and support reference-based scoring. We use a 0–100 reporting scale for novelty and embedding similarity throughout. Cosine similarity is not a calibrated probability of semantic correctness and can, in principle, be negative.

4.1 Preservation and lexical novelty

For L2-normalized embeddings from LaBSE (E_L) and GATE-AraBERT (E_G), source preservation is

$$F(P, S) = 50 [\cos(E_L(P), E_L(S)) + \cos(E_G(P), E_G(S))]. \quad (1)$$

Band membership uses this per-sentence mean, not a corpus average. Averaging encoders reduces de-

pendence on one representation but does not turn semantic similarity into an entailment or grammaticality test. We also retain reference similarity to T as a separate quantity: an identity prediction can resemble a reference that paraphrases the same source.

Let $n(x)$ remove Arabic diacritics and tatweel, normalize selected letters, remove punctuation, and split on whitespace. Letter mappings are $\bar{\text{ا}} \rightarrow \text{ا}$, $\bar{\text{ا}} \rightarrow \text{ا}$, $\text{ى} \rightarrow \text{ي}$, $\bar{\text{ه}} \rightarrow \text{ه}$, $\bar{\text{و}} \rightarrow \text{و}$, and $\bar{\text{ي}} \rightarrow \text{ي}$. The set $U(x) = \text{types}(n(x))$ contains the distinct resulting word forms. For a nonempty prediction,

$$N(P, S) = 100 \left(1 - \frac{|U(P) \cap U(S)|}{|U(P)|} \right). \quad (2)$$

Copy ratio is $100 - N$. “Types” refers to this sentence-specific set, not an inventory of grammatical categories. The asymmetric denominator ignores source words omitted from P , and repeating a type does not alter the set. Novelty therefore measures lexical change, not information retention or syntactic restructuring. Appendix B gives a worked example and invalid-output considerations.

Punctuation correction. The original normalizer (v1) preserved the Arabic Unicode block in its punctuation filter. As a result, Arabic punctuation attached to a token could create a spurious new type. The corrected normalizer (v2) additionally removes the eight punctuation characters specified in Appendix B. We rescored every system’s saved full-test outputs with v2; embedding preservation is unchanged. Corrected results are primary, while response-period experiments still using v1 are labeled explicitly. These lexical normalizations are not applied to fine-tuning inputs.

4.2 Fixed and gold-derived operating bands

The original fixed operating band is

$$\mathcal{B}_{\text{fixed}} = \{P : 40 \leq N(P, S) \leq 70, \quad F(P, S) \geq 85\}. \quad (3)$$

Coverage is the percentage of outputs satisfying both conditions. Under v2, only 46.65% of gold rewrites are inside this band. We therefore interpret it as a fixed compliance target, not a “human-quality band.” Higher coverage can suppress valid human variation.

A second, descriptive band uses the tenth and ninetieth percentiles of test-gold novelty and the

tenth percentile of test-gold preservation:

$$\mathcal{B}_{\text{gold}} = \{P : 33.33 \leq N(P, S) \leq 73.68, \quad F(P, S) \geq 76.18\}. \quad (4)$$

The percentile bounds are reported to two decimal places. Gold coverage is 74.65%, since marginal percentile conditions do not imply the same joint coverage. Both band scores are shown in the main table. Because Eq. 4 uses test references, it is a post-hoc distributional diagnostic, not a development-calibrated deployment criterion. We do not claim that this reference contact is eliminated by calling the band descriptive.

Historical diagnostic flags use $N < 25$ for copy-prone outputs, $F < 80$ for drift, and $N > 80$ for over-rewriting. These names denote threshold flags, not verified human error labels. They overlap and do not partition all out-of-band cases. No selection filter is applied to the evaluation denominator: every system’s final output is scored, including fallback outputs.

4.3 Independent and complementary measurements

We evaluate preservation with three encoders not used by the reranker, each using its own test-gold tenth-percentile floor. Alternative novelty units include light stems, root stems, and character 4-grams. These checks test sensitivity to the measurement representation; they do not constitute a gold-standard lemma-level analysis.

We also compare the joint (N, F) distributions through energy distance, with each coordinate scaled by the corresponding gold standard deviation. This distinguishes concentration inside a band from resemblance to the full annotation distribution. Its exact estimator and sampling are specified in Appendix G.

BLEU, chrF++, and ROUGE-L against T remain secondary reference-overlap measurements, with sacreBLEU where applicable (Post, 2018). ROUGE for reporting uses Arabic-normalized whitespace tokens without English stemming. High target BLEU does not inherently indicate copying, and low BLEU does not establish good paraphrasing. Independent five-point human ratings of meaning and paraphrase quality supply the most direct assessment of perceived output quality.

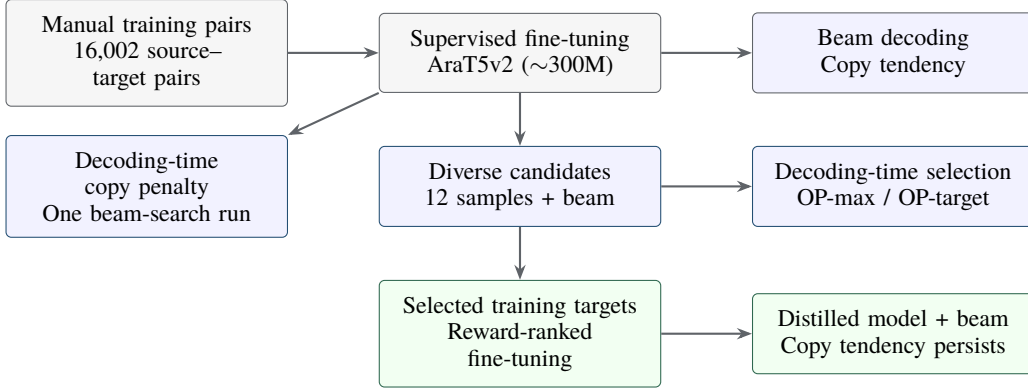


Figure 1: Decoding-time controls (blue) and reward-ranked fine-tuning (green) are separate routes. OP-max favors greater novelty; OP-target favors proximity to a chosen novelty level. Distillation uses training sources only. Human evaluation is independent of these optimization routes.

5 Methods

5.1 Base fine-tuning & candidate generation

The base model is UBC-NLP/AraT5v2-base-1024. We fine-tune for five epochs at learning rate 5×10^{-5} , with per-device train/evaluation batch size 8, label smoothing 0.1, weight decay 0.01, warm-up ratio 0.1, and fp16 precision. Maximum source and target lengths are 768 and 512 model tokens. Inputs retain their orthography and use paraphrase: {domain}: {text}; an ablation omits the domain.

The original run selected its checkpoint using development eval_rougeL computed by an English-oriented tokenizer and Porter-stemmer configuration. That criterion is unsuitable for Arabic and is a limitation of model selection. All reported evaluation scores are computed separately, but correct rescoring does not undo the possible effect of selecting a suboptimal checkpoint. We evaluate the resulting checkpoint transparently rather than imply that it was selected with the corrected metric.

Beam decoding uses five beams, no-repeat-trigram blocking, and repetition penalty 1.2. Diverse generation adds 12 nucleus samples with temperature 1.1 and top- $p = 0.95$, using the same repetition controls. Outputs are deduplicated while preserving their order: sampled candidates first, deterministic beam output last. The maximum generation length is 512 new tokens. Figure 1 places this inference procedure separately from further fine-tuning.

5.2 Feasibility, selection, and fallback

For a source S , let $t(x)$ be its normalized whitespace-token sequence and

$$\ell(P, S) = \frac{|t(P)|}{\max(1, |t(S)|)}. \quad (5)$$

The recorded selection runs use v1 lexical features; the primary table applies v2 only when evaluating their saved outputs. This is a normalized word-count ratio, not a SentencePiece length ratio. A candidate is degenerate if it has fewer than three tokens, any adjacent equal tokens, or type-token ratio below 0.55. The initial feasible set is

$$\mathcal{F}(S) = \{P : F(P, S) \geq 88, \\ 0.6 \leq \ell(P, S) \leq 1.5, \\ P \text{ is non-degenerate}\}. \quad (6)$$

The preservation floor was selected on development data from {80, 82, 84, 85, 86, 88, 90} and then fixed.

The original rule, OP-max, selects

$$P_{\max}^* = \arg \max_{P \in \mathcal{F}(S)} N(P, S). \quad (7)$$

It is useful for under-rewriting but does not enforce either novelty boundary of the evaluation band. It also shares its scoring functions with evaluation, giving it an explicit metric advantage.

The direction-neutral alternative, OP-target, selects

$$P_{\text{target}}^* = \arg \min_{P \in \mathcal{F}(S)} |N(P, S) - N^*|. \quad (8)$$

The response-period experiment uses $N^* = 56.1$, the v1 test-gold mean. This target was not

development-derived. Consequently, OP-target is evaluated as a post-hoc analysis and is not substituted for the corrected OP-max full-test result. Its exact target also cannot be carried unchanged across normalizers and called the same human operating point: v2 gold mean novelty is 52.86.

For either rule, an empty feasible set triggers the same cascade: remove the preservation floor; then remove the length window; finally, if all candidates are degenerate, return the candidate with maximum preservation over the entire pool. Retain the stated selection objective at the first nonempty stage. Ties in OP-target are resolved by the first candidate in generation order, as implemented; no preservation-based secondary tie-breaker is added retrospectively.

5.3 Fluency-constrained selection

A diagnostic extension adds a fluency floor to the feasible set. The scorer is ALLaM-AI/ALLaM-7B-Instruct-preview, run in bf16 with batch size 16 and truncation to 128 tokens. It assigns mean next-token log-likelihood over non-padding prediction positions. Higher scores indicate greater likelihood under this model, not guaranteed academic quality.

The threshold, -4.4344 , is the tenth percentile of the 2,000 test-gold scores (mean -3.6022). It is therefore another test-derived constraint. When the feasible set is empty, fluency is relaxed first, followed by preservation, length, and degeneracy. Selected outputs can consequently remain below the fluency floor. This extension is reported as an automatic diagnostic; its outputs have not received a new human audit.

5.4 Copy penalty and reward-ranked fine-tuning

For generation logit $z_t(v)$ and token-ID set V_x from the tokenized input, the copy penalty is

$$\tilde{z}_t(v) = z_t(v) - 1.5 \mathbf{1}[v \in V_x]. \quad (9)$$

It is a soft bias at every decoding step, with no decay or stop-list. The original implementation takes V_x from the *full encoded prompt*: source text, task prefix, domain, and special tokens. It thus penalizes more than source content, including function words, clitics, and punctuation. A separate ablation constructs a source-only mask without prefix or special tokens and tests function-word exemption. These are distinct implementations and their

results remain separate. The penalty value was chosen on development data. One beam-search run is required, rather than a sampled pool and two encoder scorers; measured latency or cost is not reported.

Reward-ranked fine-tuning generates and ranks candidates on 8,000 training sources. Candidate targets are retained for 82% of examples; otherwise the gold rewrite supplies the target. The base model is fine-tuned by cross-entropy using these targets (Dong et al., 2023), with the ablation/RAFT trainer using learning rate 3×10^{-5} and three epochs. This is distinct from the five-epoch primary fine-tuning run. Failure under beam decoding tests this configuration, not the general ability of cross-entropy or distillation to learn novelty control. A semantic loss computed from hard arg-max IDs would require a gradient estimator or relaxation to affect parameters; we do not present an unreported dual-loss system.

6 Experimental Protocol

Systems and snapshots. The primary comparison uses all 2,000 held-out pairs. Baselines are unadapted AraT5v2, GPT-4o, and Claude Sonnet 4.6; an additional response-period experiment evaluates ALLaM-7B-Instruct. The unadapted AraT5v2 is not instruction-tuned, so its poor behavior under a long Arabic instruction cannot establish a general failure of off-the-shelf Arabic models. Commercial identifiers are gpt-4o-2024-08-06 and claude-sonnet-4-6, queried on 18 June 2026. Original LLM generation uses temperature 0.7 and a 1,024-token output limit. Appendix C provides the original Arabic instruction and the calibrated control’s constraints.

Frontier control. The best-of-eight GPT-4o experiment uses 498 test sources: 166 evenly stepped examples from each source-length tertile. The requested sample size of 500 became 498 through integer division; no examples were lost to failed calls. Both generators are evaluated on exactly this intersection of source strings. GPT-4o receives the calibrated instruction and produces eight candidates per source at temperature 0.9. AraT5v2 uses its ordinary task/domain prefix and the existing sampled pool, averaging 12.92 candidates after deduplication. The shared selector thus reduces objective asymmetry, but prompt information and candidate budgets remain different. Neither direction of advantage justifies calling this a fully matched

comparison.

Protocol separation. The primary table uses corrected v2 novelty. The 498-source control, fluency experiments, held-out encoder results, and sampling-seed analyses retain their original v1 scores and explicit test-reference contact. We do not merge them into a new corrected leaderboard or treat them as prospectively specified tests. All final outputs remain in each evaluation denominator, regardless of selection-filter compliance.

Local computation uses one NVIDIA RTX 5090 with 32 GB memory; commercial baselines use hosted services. Three candidate-pool regenerations measure decoding variability for one fine-tuned checkpoint, not training-seed variance. The model repository is linked here.¹ Exact generation settings and supplementary scoring details appear in the appendices. Encoder revision hashes were not pinned in the recorded runs, which limits exact replication as repositories evolve.

7 Results

7.1 Corrected full-test calibration

Table 2 replaces the original full-test lexical results with the corrected scores. OP-max reaches 74.25% fixed-band coverage, compared with GPT-4o’s 52.60%, a difference of 21.65 percentage points. Under the descriptive gold-derived band, the respective rates are 85.95% and 74.25%; gold itself reaches 74.65%. Neither criterion establishes a human-quality ranking.

The correction changes the magnitude of the original result. OP-max’s novelty falls from 55.70 to 51.87, and fixed-band coverage from 76.65% to 74.25%. GPT-4o’s novelty falls from 65.14 to 62.81, bringing more outputs under the fixed upper boundary and increasing coverage from 49.35% to 52.60%. Gold novelty falls from 56.06 to 52.86. The normalization sensitivity must therefore be propagated into the gold reference as well as system scores. Appendix D reports every change.

The correction evaluates the same generated strings under a repaired measurement. Using v2 inside candidate screening and selection could also change which outputs are chosen and would require a separate reranking experiment. We make no claim for that unmeasured variant.

Fine-tuned beam outputs remain copy-prone: corrected novelty is 21.72% with domain tags and

21.71% without, with fixed coverage 5.20% and 4.95%. RAFT reduces novelty further to 18.95% and coverage to 3.30%. By contrast, the unadapted model’s high nominal novelty is accompanied by preservation near 11 and length ratio 0.18. None of its 2,000 outputs contains the affected Arabic punctuation, so its unchanged novelty after correction is compatible with degeneration rather than evidence of successful rewriting.

The copy penalty reaches 55.55% novelty and 54.45% fixed coverage with no encoder-based selection at inference. OP-max is more concentrated inside the band and has mean novelty close to the corrected gold mean. This shows that the sampled pool contains useful alternatives to the beam output. It does not establish that the model’s remaining quality limitations are solely a decoding artifact.

7.2 Calibrated frontier controls

The earlier instruction-only control reports GPT-4o coverage improving from 50.2% to 61.4% on these sources. Table 3 separately reports selection within the temperature-0.9 candidate pools; its single-candidate value is not the same generation as that instruction-only run.

OP-max is poorly suited to the calibrated GPT-4o pool: maximizing novelty yields 47.99% coverage, compared with 81.33% for target matching. AraT5v2 improves from 78.71% to 85.34% under the same change, leaving a 4.01-point gap. This supports a direction-neutral selection objective over novelty maximization when generators differ in how much they rewrite.

However, the comparison is post-hoc and not budget-matched. AraT5v2 has more candidates, while GPT-4o receives explicit novelty and length instructions. Both selectors use a target derived from test-gold v1 novelty. The result cannot establish compact-model superiority under equal computation or a fully unseen calibration protocol. It does show that supplying a stronger frontier control substantially changes the original interpretation.

7.3 Independent human quality assessment

A native Arabic PhD holder, independent of corpus construction and model development, rated the three systems on 200 test instances. Outputs were evaluated against the gold academic rewrite, with the source available as context. Two five-point ordinal dimensions assess meaning preservation and

¹<https://huggingface.co/Olmartificial-Intelligence-Space/AraScholar>

System	Preservation		Novelty v2 (%)	Length ratio	In-band (%)	
	LaBSE	GATE			Fixed	Gold-derived
Human reference	85.7	87.8	52.86	1.00	46.65	74.65
AraT5v2, zero-shot	11.4	10.4	75.46	0.18	0.00	0.00
GPT-4o, zero-shot	86.3	86.5	62.81	1.14	52.60	74.25
Claude Sonnet 4.6	82.1	83.0	73.76	1.46	22.15	41.90
AraT5v2, beam + domain	93.5	94.2	21.72	0.93	5.20	16.05
AraT5v2, beam, no domain	93.4	93.9	21.71	0.93	4.95	16.00
AraT5v2, RAFT + beam	94.3	94.6	18.95	0.93	3.30	9.60
AraT5v2, copy penalty	86.7	87.4	55.55	0.95	54.45	77.70
AraT5v2, OP-max	88.1	90.5	51.87	0.97	74.25	85.95

Table 2: Results on 2,000 test sources with corrected v2 novelty and coverage; preservation and length are unchanged. Coverage uses the fixed band or descriptive test-gold quantiles. Human scores compare gold with source; bold marks highest system coverage, not quality or significance. Copy penalty uses the original prefix-inclusive mask.

Selection on 498 sources	GPT-4o	AraT5v2
Single candidate	55.22	44.38
OP-max	47.99	78.71
OP-target	81.33	85.34
Band-membership oracle	86.14	91.16

Table 3: Post-hoc control on shared sources (v1, fixed band). GPT-4o uses calibrated prompts and eight candidates; AraT5v2 uses its ordinary prefix and 12.92 on average. OP-target uses test-gold novelty (56.1); oracle coverage indicates an in-band candidate exists.

System	Mean rating		Rating ≤ 2	
	Meaning	Para.	Meaning	Para.
AraT5v2, OP-max	3.45	2.82	33	69
GPT-4o	4.29	3.80	6	4
Claude Sonnet 4.6	4.38	3.71	6	6

Table 4: Independent single-rater audit on 200 test instances per system. Means use five-point scales; low-rating counts are out of 200. The audited small-model outputs are OP-max, not OP-target or the fluency-constrained variant.

paraphrasing quality; scores of at most two are flagged as severe.

The frontier models receive higher ratings on both dimensions (Table 4). OP-max has 69 low paraphrase-quality ratings, compared with four for GPT-4o and six for Claude. Its higher automatic calibration therefore does not translate into superior perceived writing quality in this audit. The discrepancy is a main finding, not a minor exception to the automatic comparison.

The audit’s sampling rule, identity blinding, presentation randomization, and verbatim scale anchors were not recorded in the available protocol. We do not infer them from a separate evaluation-sheet script covering a different experiment. One

Held-out encoder	Gold	GPT-4o	OP-max
Multilingual E5	75.00	71.65	85.95
Multilingual MiniLM	74.05	73.80	84.80
Arabic Triplet V2	74.60	71.15	86.25

Table 5: Coverage (%) with held-out encoders, v1 novelty bounds [35.71, 75.77], and encoder-specific test-gold P10 floors. Checkpoints and thresholds are in Appendix E.

rater provides no inter-rater reliability estimate, and a visible reference may anchor judgments toward a particular acceptable wording. These limits qualify the audit without erasing the observed quality gap.

7.4 Fluency and evaluation robustness

The OP-max selector can choose semantically similar but poorly formed candidates. Consistent with this mechanism, 44.85% of its outputs fall below the test-gold tenth-percentile LM score. Under v1, target matching with the fluency floor lowers this fraction to 15.30%, with 74.05% fixed-band coverage and 0.05% over-rewriting. Gold’s below-floor rate is 10% by construction. The selector relaxes fluency when necessary, explaining why the final below-floor rate exceeds zero.

The full ablation in Appendix F shows a trade-off: unfiltered target matching reaches 83.90% full-test fixed coverage but leaves 43.35% below the fluency threshold. Adding fluency improves that proxy at the expense of band coverage. No corresponding human re-rating has been completed, so neither human-quality improvement nor closure of the frontier gap is claimed. The 498-source 85.34% and full-test 83.90% are separate denominators, not conflicting estimates of the same run.

The held-out encoders retain OP-max’s coverage advantage (Table 5). This weakens the explanation that the result depends uniquely on the two selection encoders. It does not eliminate metric dependence: these analyses still use related embedding similarities, lexical novelty, and test-gold thresholds. Rankings also persist with light-stem, root-stem, and character-4-gram novelty; no numerical lemma-level result is claimed.

Energy distance yields a different ordering: the copy penalty is closest to gold (0.0163), followed by OP-max (0.0992), GPT-4o (0.2139), and Claude (0.9373). Independent gold subsamples give 0.0017 as a finite-sample reference. Thus, OP-max leads on band coverage while the copy penalty better matches the distribution under this statistic. The complete estimator and results are retained in Appendix G.

7.5 Additional controls and negative results

Exempting function words from the source-only copy penalty yields 56.45% v1 coverage and improves preservation from 86.97 to 88.76. This ablation differs from the prefix-inclusive primary implementation: the two all-token variants produce different outputs on 296 of 2,000 inputs (Appendix H). Across three candidate-pool regenerations, OP-max v1 coverage ranges from 77.15% to 77.75% (population SD 0.27 points); the original 76.65% comes from a separate run. These results characterize limited decoding variability, not fine-tuning variance.

Domain tags have little effect, increasing corrected beam coverage from 4.95% to 5.20%. Across 36 domains with at least ten examples, exploratory v1 OP-max coverage averages 75.2% (between-domain SD 12.0 points), without establishing uniform performance across all labels. ALLaM-7B-Instruct achieves 29.7% v1 coverage, extending the Arabic-specific baselines. Finally, RAFT lowers validation loss but reduces corrected coverage from 5.20% to 3.30%. Thus, the tested distillation configuration does not resolve beam-search copying; this finding does not rule out better targets, alternative objectives, or other decoding strategies.

8 Discussion

What calibration establishes. Joint novelty and preservation reveal failures that either axis alone can hide. The corrected results demonstrate con-

trol of measurable rewriting behavior, while gold’s low fixed-band coverage shows that a narrow rectangle cannot define paraphrase correctness. Gold-derived percentiles better describe the annotation distribution, but using test references makes that analysis retrospective. A deployment evaluation should fix its target and thresholds on development data before inspecting test outcomes.

What it does not establish. The human audit favors frontier writing even when OP-max has higher calibration scores. A fluency floor targets one missing dimension but neither proves semantic correctness nor substitutes for human re-evaluation. Similarly, a 300M generator does not make the complete reranking pipeline a 300M system: it also generates multiple candidates and invokes two encoders. Parameter count alone therefore cannot substantiate a measured cost advantage.

Examples and error interpretation. Appendix J presents two test pairs near median v1 gold novelty: neither gold rewrite meets the fixed preservation floor, and only one OP-max output is in-band, illustrating that typical human rewrites need not satisfy the band. Among 64 flagged beam outputs, 55 involve copying, supporting the metric on clear failures without establishing test-wide prevalence. Unreported fallback frequencies limit interpretation of constraint satisfaction.

9 Conclusion

AraScholar contributes 20,002 manually constructed academic paraphrase pairs and a framework for examining lexical novelty together with semantic preservation. Corrected evaluation shows that decoding controls substantially change a compact model’s operating behavior, while the tested distillation configuration remains copy-prone under beam search. Calibrated frontier controls, gold-derived diagnostics, and an independent human audit qualify the original comparison: control over automatic constraints is useful, but is not equivalent to better academic paraphrasing. Progress requires development-calibrated objectives, matched generation budgets, and independent evaluation of scientific adequacy and linguistic acceptability.

Limitations

AraScholar evaluates sentence-level rewriting and does not assess document coherence, citation in-

tegrity, or terminology consistency. Its quotation-based collection, uneven domain coverage, and annotation guidelines limit representativeness; source-level disjointness also does not establish document-level or semantic deduplication. Some annotator and quality-control details remain undocumented. Several calibration targets and thresholds use test references, making those analyses post-hoc, while supplementary v1 experiments retain the punctuation issue corrected in the main evaluation. Shared selection and evaluation metrics further limit claims about independent quality. Reproducibility is constrained by unsuitable Arabic ROUGE checkpoint selection, unpinned encoder revisions, unequal candidate budgets, and missing fine-tuning variance, significance, and inference-cost measurements. The human audit involves one evaluator with incompletely recorded procedures, and the fluency-constrained outputs have not been reassessed by humans. Hosted-model changes and restricted source access impose additional constraints.

Ethics

AraScholar supports academic rewriting with appropriate attribution: paraphrasing does not remove citation obligations or establish permission to reuse copyrighted material. Authors remain responsible for checking meaning, terminology, and unsupported claims. Each corpus item records its quoted work and containing document, but attribution alone does not establish redistribution rights. Source quotations are designated for controlled research access, without a blanket open license; the guide, code, and annotations require terms consistent with contributor rights. Model availability is separate from corpus access, and the manuscript package contains only the attributed quotations used as examples.

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A Corpus Documentation and Domains

Provenance. The field `مصدر الاقتباس` contains the quoted work’s citation, while `المصدر` identifies the containing document in which the quotation was found. Both are nonempty for every record. The 18,752 distinct quoted-work strings and 2,383 containing-document strings should not be interpreted as bibliographically deduplicated publication counts. Gregorian years parsable from the citations span 123 distinct values, with median 2002 and tenth/ninetieth percentiles 1978/2016; these summaries concern parsable years rather than every citation.

Annotation process. Collection, manual rewriting, linguistic review, and quality control were assigned to four separate team roles. The methodology requires complete quotations, changed wording with preserved meaning, cohesion, linguistic correctness, and a target no longer than its source. Review can correct errors in the printed source. The observed 61.8% length-compliance rate is reported instead of treating the guideline as an invariant. No multiple-rater agreement statistic or quantified rejection rate is available. Missing personnel and review statistics are not reconstructed from the final corpus.

Inventory. The experiment package’s `domain_inventory.json` records raw labels with split counts. Definite-article stripping yields 495 normalized labels from 522 raw forms; the mapping is mechanical, not a curated ontology. Table 6 shows the leading normalized labels. The 21 broad categories in the collection plan have no per-item mapping in the delivered corpus, so their planned allocations are not used as corpus statistics.

B Lexical Normalization

The v1 punctuation regular expression preserved characters in the Arabic Unicode block U+0600–U+06FF, including punctuation. The v2 correction additionally strips U+060C, U+061B, U+061F, U+066A, U+066B, U+066C, U+066D, and U+06D4: Arabic comma, semicolon, question mark, percent sign, decimal separator, thousands separator, five-pointed star, and full stop. Other normalization steps are unchanged. The correction rescored saved predictions; it did not regenerate or reselect candidates.

Label	English gloss	Pairs
لغة	Language	1,583
أدب	Literature	912
فلسفة	Philosophy	858
قانون	Law	851
أدب عربي	Arabic literature	807
بلاغة	Rhetoric	700
علم اجتماع	Sociology	647
اجتماع	Society / sociology	629
لسانيات	Linguistics	548
نحو	Grammar	501

Table 6: Leading corpus labels after definite-article stripping. Related categories remain separate because no manual semantic consolidation was performed.

For an illustrative source `يدرس الطالب اللغة العربية` (“The student studies the Arabic language”) and prediction `يتعلم الطالب العربية` (“The student learns Arabic”), the normalized type sets are

$$U(S) = \{\text{يدرس, الطالب, اللغة, العربية}\}, \\ U(P) = \{\text{يتعلم, الطالب, العربية}\}.$$

Two of the prediction’s three types are shared, giving copy ratio 66.67% and novelty 33.33%. Repeating a word does not change these sets, motivating separate degeneracy checks. These sentences explain the metric and are not presented as empirical dataset examples.

Empty normalized predictions have no nonzero denominator in Eq. 2. They must be treated as invalid rather than rewarded as successful novelty. The selection filter rejects outputs with fewer than three normalized tokens before normal selection; its all-degenerate fallback is still evaluated without discarding the output. Lexical normalization does not remove all morphological variation, and the light-stem, root-stem, and character-4-gram checks are distinct measurements rather than full lemmatization.

C Generation and Prompts

The source prefix is `paraphrase: {domain}: {text}`, with no text normalization. The domain ablation omits the domain component. The original instruction for frontier baselines is:

أعد صياغة النص التالي بأسلوب أكاديمي رصين مع الحفاظ
التام على المعنى، وتغيير التركيب اللغوي والمفردات قدر الإمكان
دون إضافة أي معلومات جديدة أو حذف معلومات أساسية.
اكتب النص المعاد صياغته فقط، دون أي مقدمات أو شروح.

Setting	Value
Base checkpoint	UBC-NLP/ AraT5v2-base-1024
Base fine-tuning	5 epochs; learning rate 5×10^{-5}
Per-device batch	train/eval 8 / 8
Label smoothing	0.1
Weight decay	0.01
Warm-up ratio	0.1
Precision	fp16
Maximum source/target	768 / 512 model tokens
Beam decoding	5 beams; no-repeat 3-gram; repetition penalty 1.2
Sampling	12 samples; temperature 1.1; top- p 0.95
Generation limit	512 new tokens
Original LLM calls	Temperature 0.7; 1,024-token limit
GPT-4o candidate pools	Temperature 0.9; 8 samples per call
RAFT adaptation	3 epochs; learning rate 3×10^{-5}
Local hardware	One RTX 5090, 32 GB

Table 7: Recorded generation and fine-tuning settings. The primary checkpoint’s selection criterion was the unsuitable English-oriented ROUGE configuration disclosed in Section 5.

Its English gloss is: “Rewrite the following text in a rigorous academic style while fully preserving meaning, changing structure and vocabulary as much as possible without adding information or deleting essential information. Output only the rewritten text, without introductions or explanations.”

The calibrated prompt explicitly asks for replacement of approximately 55–60% of words, retention of 40–45% (especially technical terms and proper nouns), and output length 0.9–1.2 times the source. This natural-language word-replacement instruction approximates the intended operating point; it is not an exact specification of output-type novelty. The archived CALIB_PROMPT in `rebuttal_llm_control.py` is the verbatim control prompt; the constraints above describe its calibration targets.

The primary checkpoint was produced by `train_arat5_nonnormalized.py`. The separate `train_sft.py` implements the ablation/RAFT settings. These scripts must not be conflated when reproducing the model. The primary selection criterion used `RougeScorer(use_stemmer=True)`, including English-oriented tokenization, which yields an unreliable Arabic signal. Reporting-time scores

are computed separately by `score_all.py`. Correct scoring does not establish that the best available training checkpoint was chosen.

D Normalization and Band Sensitivity

Table 8 gives the full normalization comparison. All systems are rescored on their existing 2,000 outputs. The original fixed band uses novelty [40, 70] and preservation at least 85. Test-gold percentile bands under v1 are:

- P10–P90 novelty [35.71, 75.77], P10 preservation 76.18;
- P5–P95 novelty [30.43, 80.95], P5 preservation 70.80;
- P25–P75 novelty [45.45, 66.67], P25 preservation 83.20.

Under v2, the first interval becomes [33.33, 73.68], with preservation floor unchanged at 76.18. These are test-distribution summaries; no development-derived percentile headline is claimed. Rounded means can make a difference computed from displayed cells differ by 0.01 from an underlying unrounded difference, so the table reports the original per-system values rather than reconstructed deltas.

The primary rerank floor 88 and copy penalty 1.5 were fixed on development data. By contrast, the later OP-target novelty value and fluency threshold were obtained from test gold and actively used for candidate selection. This is stronger test-reference contact than computing a descriptive percentile band alone and is explicitly disclosed. A subsequent development-calibrated evaluation should estimate those values independently and regenerate the relevant selections under v2.

E Held-out Preservation Encoders

The held-out models in Table 9 use L2-normalized sentence embeddings and prediction-to-source cosine. Inputs are truncated to 1,200 characters and the encoder’s maximum sequence length is capped at 256. The novelty interval remains the v1 test-gold range [35.71, 75.77]. The five encoder repositories were not pinned to revision hashes in the recorded runs, so exact historical weight versions cannot be established from their identifiers alone.

Additional coverage values for the copy penalty are 78.25%, 76.40%, and 76.05% under E5, MiniLM, and Arabic Triplet V2, respectively. Claude’s values are 40.35%, 43.95%, and 41.05%.

System	Novelty		Fixed-band coverage		Gold-band coverage	
	v1	v2	v1	v2	v1	v2
Human reference	56.06	52.86	50.60	46.65	75.25	74.65
OP-max	55.70	51.87	76.65	74.25	86.15	85.95
Copy penalty	58.21	55.55	55.90	54.45	77.70	77.70
GPT-4o	65.14	62.81	49.35	52.60	74.35	74.25
Claude Sonnet 4.6	75.40	73.76	18.10	22.15	43.20	41.90
Beam + domain	25.56	21.72	9.95	5.20	18.45	16.05
Beam, no domain	25.53	21.71	9.55	4.95	—	16.00
RAFT + beam	23.10	18.95	6.40	3.30	12.20	9.60
AraT5v2, zero-shot	75.46	75.46	0.00	0.00	0.00	0.00

Table 8: Full-test metric sensitivity (all values on a 0–100 scale). The fixed band is unchanged between normalizers. Each gold-derived band instead uses that normalizer’s test-gold P10/P90 novelty and P10 preservation. A dash denotes an unreported supplementary value.

Encoder	Checkpoint identifier	Gold mean	Gold P10 floor
Selection: LaBSE	sentence-transformers/LaBSE	—	—
Selection: GATE	Omartificial-Intelligence-Space/GATE-AraBert-v1	—	—
E5	intfloat/multilingual-e5-base	94.84	91.50
MiniLM	sentence-transformers/ paraphrase-multilingual-MiniLM-L12-v2	87.17	73.68
Arabic Triplet V2	Omartificial-Intelligence-Space/ Arabic-Triplet-Matryoshka-V2	88.85	78.31

Table 9: Selection and held-out encoder identifiers. LaBSE and GATE use a joint mean preservation floor of 88. The other encoders use their individual test-gold P10 floors and do not select the evaluated predictions.

OP-max remains highest in this comparison. A planned evaluation with Alibaba-NLP/gte-multilingual-base could not complete because of a runtime failure and is not included; its missing result is not treated as a negative score.

F Fluency Ablation

For an output token sequence $w_1, \dots, w_m$, the fluency score averages realized next-token log probabilities over scored non-padding positions:

$$A(P) = \frac{1}{M} \sum_{t \in \mathcal{I}} \log p(w_t \mid w_{<t}), \quad (10)$$

where $\mathcal{I}$ contains the M scored positions after truncation. The model is used as a likelihood judge in this experiment; its separate generation-baseline result is a different use. No human feedback is incorporated into candidate selection.

The feasibility cascade is: all conditions including fluency; then omit fluency; then omit preservation; then omit length; finally maximize preservation if all candidates are degenerate. Relaxing fluency explains the remaining 15.30% below-floor outputs. The 74.05% coverage checks only novelty and preservation, not a three-way acceptability conjunction. The scoring model, test-derived

threshold, 128-token truncation, and absence of a human re-audit all limit interpretation.

G Distributional Distance

Let X and Y be system and gold samples of the two-dimensional (N, F) vector, after dividing the coordinates by gold standard deviations 15.4821 and 9.1081 under v1. The same divisors are used for every system. The energy statistic is estimated as

$$\begin{aligned} \hat{\mathcal{E}}(X, Y) = & \frac{2}{mn} \sum_{i,j} \|x_i - y_j\|_2 \\ & - \frac{1}{m^2} \sum_{i,i'} \|x_i - x_{i'}\|_2 \\ & - \frac{1}{n^2} \sum_{j,j'} \|y_j - y_{j'}\|_2. \end{aligned} \quad (11)$$

For each distribution, 1,200 of the 2,000 points are sampled without replacement using NumPy’s default random generator with seed 0. Full pairwise-distance means are computed on these subsamples. Gold compared with an independent gold subsample yields 0.0017: a finite-sample sanity check, not a theoretical nonzero lower bound.

The copy penalty is closest to gold, despite OP-max’s higher band coverage. This inversion is in-

Selector	In-band	Novelty	Preservation	Below fluency floor	Over-rewrite
OP-max	76.65	55.70	89.30	44.85	2.75
OP-max + fluency	69.40	51.44	89.83	15.85	2.60
OP-target	83.90	50.98	90.39	43.35	0.00
OP-target + fluency	74.05	47.99	90.80	15.30	0.05
Human reference	50.60	56.06	86.74	10.00	18.15

Table 10: Full-test post-hoc fluency experiment using v1 novelty and the original fixed band. Fluency is the ALLaM mean token log-likelihood; its threshold is test-gold P10 (-4.4344). All entries are scores or percentages on a 0–100 scale except the threshold itself. These outputs were not re-rated by the human evaluator.

System	Energy statistic
Gold subsample vs. gold subsample	0.0017
Copy penalty	0.0163
OP-max	0.0992
GPT-4o	0.2139
Claude Sonnet 4.6	0.9373
AraT5v2, beam	2.0380
RAFT + beam	2.3640
AraT5v2, zero-shot	13.4994

Table 11: Distributional comparison under v1 with shared gold-standard-deviation scaling. Lower values indicate greater similarity under this statistic, not superior sentence quality.

Seed	In-band	Novelty	Preservation
1	77.15	55.80	89.04
2	77.75	55.94	89.25
3	77.20	56.18	89.17
Mean	77.37	55.97	89.15
Population SD	0.27	0.16	0.09

Table 12: Three full pool regenerations on 2,000 sources: v1, original fixed band, OP-max with floor 88. SD uses $\text{ddof} = 0$; it is neither a standard error nor a confidence interval.

formative: the measurements distinguish a distribution’s overall shape from its concentration inside a chosen acceptance region. These distances must not be compared numerically with values computed under another normalizer, coordinate scaling, or estimator.

H Sampling, Penalty Masks, and Diagnostics

Both PyTorch and NumPy seeds are set for each regeneration. Sampling uses 12 candidates at temperature 1.1 and top- p 0.95, no-repeat-trigram blocking, repetition penalty 1.2, plus the deterministic beam-5 output, with order-preserving deduplication. The 76.65% original result is a separate generation, not one of the three runs in Table 12. Its difference from the three-run mean is reported

without establishing sampling as the sole possible cause.

The original copy-penalty script uses the token IDs of the full encoded prompt, including prefix and special tokens, and generates one sentence at a time. The source-only ablation tokenizes the source without prefix or special tokens and runs batches of 16 with padding. Thus, both the token mask and execution configuration differ. Of the 2,000 predictions, 1,704 are identical and 296 differ; the difference cannot be uniquely attributed to the mask without holding batching fixed.

The original all-token run has 55.90% v1 fixed coverage, 14.0% drift, and 7.2% over-rewriting. The source-only all-token ablation gives 55.80%, 14.2%, and 7.3%, respectively. Exempting function words yields 56.45% coverage, novelty 51.4, preservation 88.76, drift 10.3%, and over-rewriting 3.4%. The source-only all-token preservation is 86.97. These are v1 ablation scores, not replacements for the corrected prefix-inclusive primary row.

Random selection from the original AraT5v2 pool yields 40.2% fixed coverage, versus 10.0% for beam alone and 76.65% for OP-max; its band-membership oracle reaches 91.4%. This shows a contribution from both candidate generation and selection, but failure to attain the oracle does not remove shared-metric bias. The post-hoc test sweep reaches 79.3% at preservation floor 86; the reported primary setting remains the development-selected 88.

A native-speaker inspection of 64 flagged base-beam outputs labels 53 source reproductions, two trivial edits, four degenerate repetitions, three meaning/negation/deletion errors, and two simplified-register cases. Copying-related cases total 55/64 (85.9%), with mean copy ratio 98.5% and median 100%. This selected subset supports interpretation of clear copying but provides neither full-test error prevalence nor validation of all band

System	BLEU	chrF++	ROUGE-L
AraT5v2, zero-shot	0.0	2.5	4.5
GPT-4o	13.4	44.4	35.5
Claude Sonnet 4.6	5.7	37.3	25.8
Beam + domain	20.6	47.9	44.2
Beam, no domain	20.3	47.9	44.0
RAFT + beam	19.0	47.3	43.4
Copy penalty	15.2	42.0	35.7
OP-max	10.9	39.2	31.9

Table 13: Original reference-overlap measurements on 2,000 outputs, retained as historical diagnostics. They are not represented as a newly regenerated v2 overlap table.

thresholds.

I Reference-based Scores

The source-preservation scores in Table 2 compare predictions with the input. Target similarity instead compares them with the gold rewrite. Historical LaBSE/GATE target similarities are 87.0/86.9 for GPT-4o, 82.9/82.8 for Claude, 88.0/89.0 for the domain-conditioned beam model, 87.8/88.8 for RAFT, 85.1/85.3 for the copy penalty, and 84.5/86.4 for OP-max. Copy-prone models can score highly because gold and source already convey similar content. Neither target similarity nor the overlap values alone determine how well a system rewrites.

J Representative Gold-Novelty Examples

Tables 14 and 15 report actual test records near the median of the *v1 gold novelty* distribution. The examples were not selected for a favorable system ranking. Their scores retain v1, with preservation computed by Eq. 1; they must not be compared to a v2 novelty boundary without rescoring. English glosses explain the Arabic outputs and are not additional references used for evaluation.

In Example 1, gold novelty is at the 51.2nd percentile, but its preservation score is 80.1. In Example 2, gold novelty is at the 54.8th percentile, but preservation is 78.6. Both fail the fixed floor of 85. OP-max has novelty 37.5 in the first example and therefore also falls outside the fixed band; it satisfies the band in the second. These cases concretely illustrate why typical lexical change and fixed-band compliance are different properties.

The Arabic outputs also reveal limitations of lexical counts. The beam model in Example 1 copies almost exactly, while OP-max changes the opening but retains much of the structure. Claude in-

troduces “intellectual orientations,” inability to advance, and a methodological characterization, illustrating possible elaboration alongside high novelty. In Example 2, the gold target itself strengthens “cares about nothing more than” to “cares only about.” Qualitative interpretation should therefore examine semantic relations directly rather than assume that any gold or in-band sentence is beyond criticism.

Role	Arabic output and English gloss	Nov.	Pres.
Source	الإرهاصات تعبر عن نوايا نعلنها جميعاً دون أن تتحول هذه النوايا إلى علم دقيق <i>Precursors express intentions we all declare, without these intentions becoming an exact science.</i>	–	–
Gold	الإرهاصات تمثل نوايا معلنة لدى الجميع، لكنها لم تتحول بعد إلى علم دقيق ومنظم. <i>Precursors represent intentions declared by everyone, but they have not yet become a precise, organized science.</i>	57.1	80.1
Beam	الإرهاصات تعبر عن نوايا نعلنها جميعاً دون أن تتحول هذه النوايا إلى علم دقيق. <i>Precursors express intentions we all declare, without these intentions becoming an exact science.</i>	0.0	98.2
OP-max	تظهر الإرهاصات على شكل نوايا نعبّر عنها جميعاً، دون أن تتحول هذه النوايا إلى علم دقيق. <i>Precursors appear as intentions we all express, without these intentions becoming an exact science.</i>	37.5	92.3
Copy penalty	تُعرّف الإرهاصات بأنها نوايا نعلنها جميعاً، لكنها لا تصبح علماً دقيقاً. <i>Precursors are defined as intentions we all announce, but they do not become an exact science.</i>	72.7	79.4
GPT-4o	الإرهاصات تشير إلى نوايا نصرّح بها جميعاً، إلا أن هذه النوايا لا تتطور إلى معرفة دقيقة. <i>Precursors indicate intentions we all declare, but these intentions do not develop into precise knowledge.</i>	60.0	86.4
Claude	تُجسّد الإرهاصات توجهاتٍ فكريةً يُفصح عنها الجميع، غير أنها تظل عاجزةً عن الارتقاء إلى مستوى المعرفة العلمية الدقيقة والمنهجية. <i>Precursors embody intellectual orientations expressed by everyone, yet remain unable to rise to precise, methodical scientific knowledge.</i>	84.2	56.7

Table 14: Example 1: test item 18561, Philosophy. Quoted-work attribution: Hassan Hanafi, *Muqaddimah fi 'Ilm al-Istighrab*, University Institution for Studies, Publishing and Distribution, Beirut, third edition, 2006, p. 48. Scores use v1. Gold and OP-max both fall outside the fixed band, for different reasons.

Role	Arabic output and English gloss	Nov.	Pres.
Source	القصة لا تهتم بشيء أكثر من اهتمامها باستكشاف الحقائق في الأمور الصغيرة المألوفة العادية. <i>The story cares about nothing more than exploring truths in small, familiar, ordinary matters.</i>	–	–
Gold	لا تهتم القصة سوى بالكشف عن الحقائق في تفاصيل الحياة اليومية البسيطة. <i>The story is concerned only with uncovering truths in the simple details of everyday life.</i>	58.3	78.6
Beam	لا تهتم القصة بشيء أكثر من اهتمامها باستكشاف الحقائق في الأمور البسيطة والمألوفة. <i>The story cares about nothing more than exploring truths in simple, familiar matters.</i>	15.4	95.4
OP-max	لا تشغل القصة اهتماماً أكبر من البحث عن الحقائق في الأمور البسيطة المألوفة. <i>The story occupies no greater interest than the search for truths in simple, familiar matters.</i>	46.2	89.8
Copy penalty	تُركّز القصة على استكشاف الحقائق المتعلقة بالأمور البسيطة والمألوفة. <i>The story focuses on exploring truths concerning simple and familiar matters.</i>	77.8	63.8
GPT-4o	تهدف القصة بشكل أساسي إلى استكشاف الحقائق الكامنة في التفاصيل الدقيقة للأمور اليومية المألوفة. <i>The story primarily aims to explore truths underlying the fine details of familiar everyday matters.</i>	71.4	58.4
Claude	تنصبُّ عناية القصة في جوهرها على استجلاء الحقائق الكامنة في تفاصيل الحياة اليومية المعتادة والمواقف الاعتيادية المألوفة. <i>The story's essential concern centers on elucidating truths in the details of ordinary daily life and familiar, routine situations.</i>	75.0	59.1

Table 15: Example 2: test item 10817, Arabic Literature. Quoted-work attribution: Mukhtar 'Ajubah, *Al-Qissah al-Hadithah fi al-Sudan*, Sudanese Studies Center, Cairo, second edition, 2000, p. 18. Scores use v1. OP-max satisfies the fixed band, while the gold reference falls below its preservation floor.